

Robot Task API Documentation

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Base URL: `http://ServerIP:8080`

Authentication: All endpoints in this section are anonymous and do not require a token.

Delivery Task

Request URL

- `/device/deliveryTask`

Request Method

- POST

After receiving the request, the cloud server calls the robot's RCS port 8068 to dispatch a delivery task.

Request Example

```
POST /device/deliveryTask
{
  "deviceIp": "192.168.0.120",
  "taskId": "DELIVERY-1773990764094",
  "station": "pt-1,pt-2",
  "mapName": "Ezhan1_8,Ezhan1_9"
}
```

复制



Parameters



Parameter	Required	Type	Description
deviceIp	Yes	string	Robot IP address
taskId	Yes	string	Task ID
station	Yes	string	Station name list, comma-separated
mapName	No	string	Map name list with building and floor info, comma-separated

Response

```
{
  "code": 200,
  "msg": "Delivery task dispatched",
  "taskId": "DELIVERY-1773990764094"
}
```

Cruise Task

Request URL

- `/device/cruiseTask`

Request Method

- POST

After receiving the request, the cloud server calls the robot's RCS port 8068 to dispatch a cruise task.

Request Example



```
POST /device/cruiseTask
{
  "deviceIp": "192.168.0.120",
  "taskId": "CRUISE-1773990764094",
  "taskName": "Floor 1 Cruise"
}
```

Parameters

Parameter	Required	Type	Description
deviceIp	Yes	string	Robot IP address
taskId	Yes	string	Task ID
taskName	No	string	Task name

Response

```
{
  "code": 200,
  "msg": "Cruise task dispatched",
  "taskId": "CRUISE-1773990764094"
}
```

Jack Task

Request URL

- `/device/jackTask`

Request Method

- POST

After receiving the request, the cloud server calls the robot's RCS port 8068 to dispatch a jack task.



Request Example



```
POST /device/jackTask
{
  "deviceIp": "192.168.0.120",
  "taskId": "JACK-1773990764094",
  "taskName": "Medicine Transport"
}
```

Parameters

Parameter	Required	Type	Description
deviceIp	Yes	string	Robot IP address
taskId	Yes	string	Task ID
taskName	No	string	Task name

Response

```
{
  "code": 200,
  "msg": "Jack task dispatched",
  "taskId": "JACK-1773990764094"
}
```

Cancel Task

Request URL

- `/device/cancelTask` (Admin portal)
- `/device/cancelTaskForApp` (APP)

Request Method

- POST

Request Example





```
POST /device/cancelTask
{
  "deviceIp": "192.168.0.120"
}
```

Parameters

Parameter	Required	Type	Description
deviceIp	Yes	string	Robot IP address

Response



```
{
  "code": 200,
  "msg": "Cancel task dispatched"
}
```

Pause Task

Request URL

- `/device/pauseTask`

Request Method

- POST

Request Example



```
POST /device/pauseTask
{
  "deviceIp": "192.168.0.120"
}
```



Parameters



Parameter	Required	Type	Description
deviceIp	Yes	string	Robot IP address

Response

```
{
  "code": 200,
  "msg": "Pause task dispatched"
}
```

Resume Task

Request URL

- `/device/resumeTask`

Request Method

- POST

Request Example

```
POST /device/resumeTask
{
  "deviceIp": "192.168.0.120"
}
```

Parameters

Parameter	Required	Type	Description
deviceIp	Yes	string	Robot IP address

Response



```
{
  "code": 200,
  "msg": "Resume task dispatched"
}
```

Charge Task

Request URL

- `/device/chargeTask`

Request Method

- POST

Request Example

```
POST /device/chargeTask
{
  "deviceIp": "192.168.0.120"
}
```

Parameters

Parameter	Required	Type	Description
deviceIp	Yes	string	Robot IP address

Response

```
{
  "code": 200,
  "msg": "Charge task dispatched"
}
```



Query Robot Status

Request URL

- `/device/queryRobotStatus`

Request Method

- GET

Request Example

GET /device/queryRobotStatus?deviceIp=192.168.0.120

Parameters

Parameter	Required	Type	Description
deviceIp	Yes	string	Robot IP address

Response

```
{
  "code": 200,
  "msg": "Query successful",
  "data": {
    "status": 1,
    "mapName": "81",
    "x": 2.33,
    "y": 1.62,
    "theta": 2.73
  }
}
```

Response Parameters

Parameter	Type	Description
status	int	Robot status: 1=Idle, 2=Running, 3=Error, 4=Offline
mapName	string	Current map name
x	float	X coordinate



Parameter	Type	Description
y	float	Y coordinate
theta	float	Heading angle

Check Device Status (Detailed)

Request URL

- `/device/checkStatus`

Request Method

- POST

Request Example

```
POST /device/checkStatus
{
  "deviceIp": "192.168.0.120"
}
```

Parameters

Parameter	Required	Type	Description
deviceIp	Yes	string	Robot IP address

Response





```
{
  "code": 200,
  "msg": "Device online",
  "data": {
    "connected": true,
    "power": 52.95,
    "powerQuantity": 97,
    "emgStop": false,
    "collisionWarning": false,
    "movement": 0,
    "task_status": "EXECUTING",
    "taskType": "delivery",
    "taskName": "Medicine Delivery",
    "currentLocation": "Ward A",
    "mapName": "81",
    "x": 5.2,
    "y": 3.1,
    "theta": 1.57,
    "robotId": "default",
    "version": "20260113v1",
    "odomHz": 93.16,
    "scanHz": 14.91
  }
}
```

Response Parameters

Parameter	Type	Description
connected	bool	Whether online
power	float	Voltage
powerQuantity	int	Battery percentage
emgStop	bool	Emergency stop activated
collisionWarning	bool	Collision sensor triggered
movement	int	Movement status: 0=Stopped, 1=Moving
task_status	string	Task status
taskType	string	Task type



Parameter	Type	Description
taskName	string	Task name
currentLocation	string	Current station
mapName	string	Map name
x	float	X coordinate
y	float	Y coordinate
theta	float	Heading angle

Appendix

Common Response Codes

Code	Description
200	Success
500	Failure / Server error
401	Unauthorized, token invalid or expired

Device Status Enum

Status	Description
1	Idle
2	Running
3	Error
4	Offline



Task Type Enum



TaskType	Description
delivery	Delivery task
cruise	Cruise task
jack	Jack / Transport task
call	Single-point call task
park	Park task
charge	Charge task
pause	Pause task
resume	Resume task
cancel	Cancel task

